

FUAD AHMED (TUBA, FUAD AHMED)

YENCHING ACADEMY OF PEKING UNIVERSITY • DOSSIER

Host: Peking University, Beijing, China • **Fellowship:** Yenching Academy Full Fellowship

Research Track: Law and Society / Economics & Management (Digital Infrastructure & Tech Policy)

Applicant: fuadahmedt@gmail.com • +251 925 278 350 • Addis Ababa, Ethiopia

PART I: PERSONAL STATEMENT (750 WORDS MAX)

To the Admissions Committee of the Yenching Academy of Peking University,

My trajectory at the intersection of technology and development is grounded in the physical reality of Ethiopia's digital emergence. As a computer science graduate from St. Mary's University and a software systems engineer with four years of hands-on experience designing distributed architectures, I have spent my career confronting the tangible frictions of the global digital divide. In Addis Ababa, software cannot exist in an abstract computational vacuum; its reliability is fundamentally constrained by network latency, power grid reliability, and the availability of sovereign cloud infrastructure.

Over the past decade, the rapid expansion of Ethiopia's telecommunications and data infrastructure has been deeply intertwined with Chinese engineering and partnership—from the construction of the National Data Center to fiber-optic backbones deployed through the Digital Silk Road. While working as a backend engineer building high-concurrency enterprise services, I realized that true digital transformation requires more than writing efficient code. It requires an interdisciplinary understanding of how technological standards, international economic cooperation, and institutional policies shape national sovereignty and economic agency.

The Yenching Academy of Peking University represents the premier institution where I can bridge my technical foundation with advanced China Studies. Peking University's historic leadership in geopolitical thought, combined with China's unprecedented domestic scaling of cloud ecosystems and digital governance, offers an intellectual environment that cannot be replicated in the West. At Yenching Academy, I aim to engage deeply with peers across politics, economics, and international relations, analyzing how Chinese digital infrastructure models can inform sustainable, sovereign technological frameworks across Africa.

I bring to the Yenching cohort a distinct operational perspective: that of a practicing software engineer from East Africa who understands the code beneath digital trade and the human stakes of technological access. Supported by the Yenching Fellowship, I am prepared to contribute actively to Peking University's interdisciplinary dialogue.

PART II: RESEARCH PROPOSAL (1,500 WORDS MAX)

Title: Evaluating Sino-African Digital Infrastructure Partnerships: Cloud Sovereignty, Distributed Telecoms, and Data Governance in East Africa

1. Research Background & Rationale

The Digital Silk Road (DSR), launched under the Belt and Road Initiative, has become the dominant framework for telecommunications modernization across sub-Saharan Africa. In Ethiopia and neighboring East African states, Chinese technological enterprises (notably Huawei and ZTE) have financed, engineered, and deployed nationwide 4G/5G mobile infrastructure, fiber-optic networks, and Tier-3 national data centers. Concurrently, developing nations are grappling with the imperatives of "data sovereignty"—the legal and technical ability of a state to retain control over citizen data, critical financial transactions, and sovereign cloud workloads.

While Western academic commentary frequently evaluates Sino-African technological cooperation through the narrow lens of geopolitical rivalry or surveillance concerns, there is a pronounced deficit of empirical, technical research conducted from the perspective of African engineers and policymakers. How are localized data localization laws actually implemented across Chinese-built cloud infrastructures? What are the technological transfer mechanisms and technical capacity-building outcomes for indigenous African software engineers operating these systems?

2. Key Research Questions

1. How do Sino-African digital infrastructure partnerships balance physical infrastructure deployment with indigenous data sovereignty and national governance in East Africa?

2. To what extent does the adoption of Chinese telecommunications and cloud standards foster local technical autonomy versus technological dependency among local engineering ecosystems?
3. What lessons can African policymakers synthesize from China's own historical trajectory in building domestic tech sovereignty, cloud ecosystems, and data governance frameworks?

3. Methodology & Interdisciplinary Framework

This project will utilize a mixed-methods, interdisciplinary approach combining technical systems analysis with political economy:

- **Technical Architecture Audit:** Examining the architectural topology of major Sino-African infrastructure projects in Ethiopia, including data routing pathways, server virtualization protocols, and API interoperability with global standards.
- **Comparative Policy Analysis:** Evaluating China's Cybersecurity Law and Data Security Law alongside the African Union's Malabo Convention on Cyber Security and Personal Data Protection and Ethiopia's Personal Data Protection Proclamation.
- **Stakeholder Fieldwork:** Conducting qualitative semi-structured interviews with Chinese tech enterprise leads in Beijing, African diplomatic representatives, and Ethiopian telecommunications engineers.

4. Expected Contributions

This research will produce actionable policy recommendations for Sino-African tech governance, providing African governments with a nuanced blueprint for negotiating equitable infrastructure agreements while advancing academic literature at Peking University on South-South digital cooperation.

Respectfully submitted,

Fuad Ahmed (TUBA, FUAD AHMED)

BSc in Computer Science, St. Mary's University • Addis Ababa, Ethiopia • fuadahmedt@gmail.com